

Calabi–Yau quantum groups and 6d holomorphic Chern–Simons

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Abstract

Volume I of the modular Koszul duality programme introduces E_1 -chiral quantum groups as abstract algebraic-geometric objects: ordered bar complexes with spectral R -matrices, Drinfeld coproducts, and Yang–Baxter equations on configuration spaces of curves. This paper constructs them concretely from Calabi–Yau geometry.

The source is the holomorphic Chern–Simons hierarchy. At the top ($d = 6$): holomorphic Chern–Simons theory on a CY_3 -fold Y . A curve $C \subset Y$ of codimension 2 carries a defect algebra: the normal bundle $N_{C/Y}$ supplies two spectral parameters, and the defect algebra is a chiral algebra on C with quantum group structure from the normal directions. For $Y = \mathbb{C}^3$ and $C = \mathbb{C}$, the local defect gives the abelian low-spin witness $J(z)J(w) \sim \Psi_{\text{lev}}/(z-w)^2$, with $\Psi_{\text{lev}} = -\sigma_2(h_1, h_2, h_3)$ and Sugawara central charge 1. The full $\mathcal{W}_{1+\infty}$ object requires the completed Drinfeld double or Fock/evaluation data. The hierarchy $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$ is codimension-2 restriction followed by curve specialization; the ambient hCS observables are holomorphic E_3 in complex dimension 3 and holomorphic E_2 in complex dimension 2, while the curve output is E_1 .

The critical cohomological Hall algebra $\text{CoHA}(Q, W)$ of a quiver with CY_3 potential is the positive E_1 -primitive: its product is ordered Hall multiplication. The full Hopf and vertex-coproduct data require localization, a Hopf pairing, the opposite half, the Cartan sector, and completion. For the Jordan quiver: $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$. The full $\mathcal{W}_{1+\infty}$ vertex realisation requires the corresponding double/evaluation construction. The Drinfeld center recovers E_2 -braiding, closing the E_n operadic circle.

Quantum toroidal algebras $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ carry the CY condition as the parameter constraint $q_1 q_2 q_3 = 1$. The structure function $G(x; q_1, q_2, q_3) = \prod_i (1 - q_i x) / \prod_i (1 - q_i^{-1} x)$ satisfies $G(x)G(1/x) = (q_1 q_2 q_3)^2$; the identity $G(x)G(1/x) = 1$ characterizes the connected CY branch $q_1 q_2 q_3 = 1$ after the exponential/formal branch is fixed.

Wall-crossing in the Bridgeland stability manifold is Maurer–Cartan gauge equivalence in the convolution algebra of Volume I: the Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation $\Theta_\sigma \sim \Theta_{\sigma'}$. The shadow obstruction tower at $\kappa_{\text{ch}} = \Psi$ recovers the genus-1 constant-map contribution for $\mathbb{C}^3$; the all-genus virtual equality is conditional on the uniform-weight and $\Phi_3^{(\Sigma_2, C)}$ hypotheses, and the motivic equality remains conjectural in general.

Contents

I	Introduction	2
1.1	CY quantum groups as concrete E_1 -chiral quantum groups	2
1.2	The holomorphic Chern–Simons hierarchy	3
1.3	The E_n operadic circle	3
1.4	Construction layers	4

2	The holomorphic Chern–Simons hierarchy	4
2.1	6d holomorphic Chern–Simons	4
2.2	Dimensional reduction: $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$	4
2.3	The Ω -background and equivariant parameters	5
3	Codimension-2 defect algebras	5
3.1	The defect construction	5
3.2	$\mathbb{C}^3$: the low-spin defect witness	5
3.3	Defect algebras for general CY_3	6
4	The CoHA as ordered Hall input	6
4.1	Critical cohomological Hall algebras	6
4.2	The CoHA carries the positive E_1 half	7
4.3	The R -matrix and the Drinfeld coproduct	7
5	Quantum toroidal algebras and the CY condition	8
5.1	The CY parameter constraint	8
5.2	The Miki automorphism	8
5.3	The DDCA–toroidal bridge	9
6	The Drinfeld center closes the E_n circle	9
7	Shadow = $\text{GW}(\mathbb{C}^3)$ and wall-crossing	10
7.1	The shadow tower as Gromov–Witten theory	10
7.2	Wall-crossing as MC gauge equivalence	10
8	The CY_4 obstruction and the p_1-twisted double current algebra	11
8.1	The obstruction in $\pi_4(BU)$	11
8.2	The Drinfeld double presentation	11
8.3	The RTT relation through $O(\hbar^2)$	11
8.4	CY_4 twisted-supergravity analogue	12
8.5	Comparison with Maulik–Okounkov	12
8.6	p_1 -twisted stratum of $K_3 \times K_3$	12
9	Codimension-2 defects, CoHA, and the toroidal formal disk	13

I Introduction

I.1 CY quantum groups as concrete E_1 -chiral quantum groups

Volume I constructs E_1 -chiral quantum groups as abstract algebraic-geometric objects on curves. An E_1 -chiral quantum group is a tuple $(\mathcal{A}, \mu, \Delta_z, \varepsilon, \eta, S)$ satisfying five axioms (H1–H5): an E_1 -chiral algebra structure, a spectral coproduct, bialgebra compatibility, spectral coassociativity

$$(\Delta_w \otimes \text{id}) \circ \Delta_{z+w} = (\text{id} \otimes \Delta_z) \circ \Delta_w,$$

and the Hopf axiom at $z = 0$. The ordered bar $B^{\text{ord}}(A) = T^c(s^{-1}\bar{A})$ preserves the R -matrix; the symmetric bar $B^{\Sigma}(A)$ kills the Hopf structure via the lossy averaging map: in abelian and scalar families $\text{av}(r(z)) = \kappa_{\text{ch}}$, while for non-abelian affine Kac–Moody $\text{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa_{\text{ch}}$.

Two geometric inputs produce concrete E_1 -chiral quantum groups: the 6d holomorphic Chern–Simons hierarchy, which gives defect chiral algebras on codimension-2 curves, and critical CoHA, which gives the ordered E_1 -primitive graded by dimension vectors.

The E_1 stabilization theorem of the companion paper proves that for $d \geq 3$ the constructed CY chiral output on the reference curve is natively E_1 on the witnessed loci. The full Schouten–Nijenhuis bracket is nonzero; the ordered CoHA direction is the quantum-group source. The braided E_2 -structure is recovered through the Drinfeld center of the E_1 -monoidal representation category. For $d \geq 3$,

$$\pi_1(\text{Conf}_2(\mathbb{R}^d)) = \pi_1(S^{d-1}) = 0, \quad \pi_1(\text{Conf}_2(\mathbb{R}^d)/\mathfrak{S}_2) = \mathbb{Z}/2,$$

so the unordered quotient supplies only symmetric braiding.

1.2 The holomorphic Chern–Simons hierarchy

The theories of this paper form a hierarchy in real dimension:

d	Theory	Curve output	E_n level
6	hCS on $\text{CY}_3 Y$	defect on $C \subset Y$	E_1
5	partially topological CS on $\mathbb{R} \times S$	$\mathbb{R} \times C$	E_1
4	hCS on $\mathbb{C}^2$ (CWY)	line operator	E_1
3	3d HT (Volume II)	boundary algebra	E_1

The hierarchy $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$ is codimension-2 restriction at each step. Each restriction produces a chiral algebra from a gauge theory. The resulting chiral algebras are E_1 -chiral quantum groups in the sense of Volume I, with R -matrices from the normal directions and coproducts from the spectral parameters.

1.3 The E_n operadic circle

The E_n hierarchy organizes the entire programme:

$$E_3(\text{bulk}) \rightarrow E_2(\text{boundary chiral}) \rightarrow E_1(\text{bar/QG}) \rightarrow E_2(\text{Drinfeld center}) \rightarrow E_3(\text{derived center}).$$

Each arrow has precise mathematical content: $E_3 \rightarrow E_2$ is restriction to a codimension-2 defect; $E_2 \rightarrow E_1$ is the ordered bar complex; $E_1 \rightarrow E_2$ is the Drinfeld center; $E_2 \rightarrow E_3$ is higher Deligne plus topologization. The round trip closes only after the Drinfeld-centre/derived-centre comparison and the topologization package are both available. This is proved on the affine Kac–Moody non-critical locus by the Sugawara BRST primitive; for general class- $\mathcal{M}$ inputs the original-complex chain-level comparison remains open. Without that primitive the structure remains $\text{SC}^{\text{ch}, \text{top}}$.

The CY quantum groups of this paper enter the circle at the E_1 vertex: the CoHA is the ordered associative Hall input, and the Drinfeld center carries it to E_2 . The tetrahedron equation (3d analogue of YBE) fails for the Yang R -matrix at $O(\kappa_{\text{ch}}^2)$: the E_3 structure is genuinely nontrivial beyond E_2 .

1.4 Construction layers

The construction has five layers. Codimension-2 hCS restriction produces the defect chiral algebra on a curve. The local $\mathbb{C}^3$ defect identifies the Heisenberg low-spin channel and isolates the completed double/Fock data needed for $\mathcal{W}_{1+\infty}$. Critical CoHA supplies the ordered positive half. The toroidal formal disk imposes $q_1 q_2 q_3 = 1$. The Drinfeld center and the shadow tower connect the E_1 primitive to E_2 braiding and the genus-one DT/GW channel.

2 The holomorphic Chern–Simons hierarchy

2.1 6d holomorphic Chern–Simons

Let Y be a CY_3 -fold with holomorphic volume form $\Omega_Y \in H^0(Y, \omega_Y)$. The holomorphic Chern–Simons functional on Y (Witten, Costello) is

$$S_{\text{hCS}}(\mathcal{A}) = \int_Y \Omega_Y \wedge \text{tr} \left(\mathcal{A} \wedge \bar{\partial} \mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \right)$$

where $\mathcal{A} \in \Omega^{0,1}(Y, \mathfrak{g})$ is a $(0,1)$ -form valued in a Lie algebra $\mathfrak{g}$. The equations of motion $\bar{\partial} \mathcal{A} + \mathcal{A} \wedge \mathcal{A} = 0$ are the integrability condition for the holomorphic structure on the trivial G -bundle.

The theory is six-dimensional, holomorphic in all three complex directions. The BRST operator is $Q_{\text{BRST}} = \bar{\partial} + [\mathcal{A}, -]$, and the observables are the Dolbeault cohomology $H_{\bar{\partial}, \mathcal{A}}^{0,\bullet}(Y, \mathfrak{g})$.

Remark 2.1 (Relation to physical dimensions). The 6d hCS theory is the Costello–Li/BCOV open B-model holomorphic field-theory avatar of the E_3 story: the kinetic term is first-order, the path integral is one-loop exact, and the partition function is a holomorphic invariant. The 6d $(2, 0)$ Schur/BLLPR route is an independent construction. The physical content sits in the defect algebras.

2.2 Dimensional reduction: $6 \rightarrow 5 \rightarrow 4 \rightarrow 3$

The hierarchy proceeds by codimension-2 restriction at each step.

$6 \rightarrow 5$: for a CY_3 -fold of the form $Y = \mathbb{R} \times S$ (where S is a CY_2 surface), the 6d hCS theory reduces to 5d partially topological Chern–Simons on $\mathbb{R} \times S$. The topological direction $\mathbb{R}$ carries the E_1 ordering; the holomorphic surface S carries the chiral data. Restriction to $\mathbb{R} \times C$ (where $C \subset S$ is a curve) produces the ordered bar complex with one spectral parameter.

$5 \rightarrow 4$: for $S = \mathbb{C}^2$, the 5d theory reduces to 4d holomorphic Chern–Simons on $\mathbb{C}^2$: the Costello–Witten–Yamazaki theory. Line operators on $\mathbb{C} \subset \mathbb{C}^2$ produce Yangians from holomorphic gauge fields. The R -matrix comes from the normal direction to the line.

$4 \rightarrow 3$: restricting to a further codimension-2 locus produces the 3d holomorphic-topological theories of Volume II. The boundary algebra is a chiral algebra on the codimension-2 curve, and the $\text{SC}^{\text{ch}, \text{top}}$ pair $(C_{\text{ch}}^\bullet(A, A), A)$ is the boundary–bulk system.

Proposition 2.2 (Dimensional consistency). *The ambient hCS observables retain their holomorphic factorization level before curve specialization: 6d hCS on a CY_3 has ambient holomorphic E_3 observables, and 5d hCS on $\mathbb{R} \times \mathbb{C}^2$ has holomorphic E_2 observables. Projection to a defect curve is the stage that produces the ordered E_1 chiral algebra with spectral parameters from the normal directions. The E_2 -braiding is recovered through the Drinfeld center.*

2.3 The Ω -background and equivariant parameters

For the toric CY_3 -fold $Y = \mathbb{C}^3$, the three coordinate directions carry equivariant parameters $h_1, h_2, h_3 \in \mathbb{C}$ subject to the CY condition $h_1 + h_2 + h_3 = 0$. In the multiplicative parametrization $q_i = e^{h_i}$, the CY condition reads $q_1 q_2 q_3 = 1$. The identification $q_1 = q, q_2 = t^{-1}, q_3 = t/q$ recovers the standard (q, t) -parametrization of the quantum toroidal algebra.

The Ω -background parameters enter the defect algebra through the equivariant Chern–Simons functional: the gauge field on Y decomposes along the normal bundle to a defect curve $C \subset Y$, and the equivariant parameters weight the normal modes. For $C = \mathbb{C}_z$ (the first coordinate axis) inside $Y = \mathbb{C}^3$, the normal bundle is $N_{C/Y} = \mathcal{O}_C^{\oplus 2}$ with fiber coordinates (w_1, w_2) carrying equivariant weights (h_2, h_3) . The resulting defect algebra on C has OPE poles involving both normal parameters.

3 Codimension-2 defect algebras

3.1 The defect construction

A curve $C \subset Y$ in a CY_3 -fold carries a defect algebra: the 6d hCS theory restricted to a tubular neighborhood of C produces a 2d chiral algebra on C with additional structure from the normal bundle.

Construction 3.1 (Codimension-2 defect algebra). Let Y be a CY_3 -fold, $C \subset Y$ a smooth curve, and $N_{C/Y}$ the normal bundle of rank 2. The defect algebra $\mathcal{A}_C^{\text{def}}$ on C is obtained by:

1. **Restriction.** Restrict the 6d hCS gauge field $\mathcal{A} \in \Omega^{\text{ol}}(Y, \mathfrak{g})$ to the formal neighborhood $\hat{N}_{C/Y}$ of C in Y . The Taylor expansion along the normal fibers gives a collection of fields on C indexed by normal-direction monomials.
2. **BRST reduction.** The BRST operator $Q = \bar{\delta} + [\mathcal{A}, -]$ restricts to the formal neighborhood. The Q -cohomology in the fiber directions produces the generating fields of the defect algebra on C .
3. **OPE from normal modes.** The OPE on C is inherited from the 6d hCS action. The normal bundle $N_{C/Y}$ supplies two spectral parameters (from the two fiber coordinates), which enter the OPE as meromorphic functions of the insertion points on C .

Remark 3.2 (Two spectral parameters). The rank-2 normal bundle $N_{C/Y}$ gives two spectral parameters. This is the geometric origin of the two-parameter family of quantum toroidal algebras: the two normal directions to a curve inside a CY_3 -fold produce the (q, t) -deformation. For a CY_2 -fold (surface), the normal bundle has rank 1, producing one spectral parameter: the standard spectral parameter of Yangians and KZ equations.

3.2 $\mathbb{C}^3$: the low-spin defect witness

Proposition 3.3 (Low-spin defect algebra of $\mathbb{C}^3$). For $Y = \mathbb{C}^3$ with coordinates (z, w_1, w_2) and CY form $\Omega = dz \wedge dw_1 \wedge dw_2$, the codimension-2 defect algebra on $C = \mathbb{C}_z$ has a Heisenberg current

$$J(z)J(w) \sim \frac{\Psi_{\text{lev}}}{(z-w)^2}, \quad \Psi_{\text{lev}} = -\sigma_2(h_1, h_2, h_3) = \frac{h_1^2 + h_2^2 + h_3^2}{2}$$

on the Calabi–Yau branch $h_1 + h_2 + h_3 = 0$. Its Sugawara stress tensor is

$$T = \frac{1}{2\Psi_{\text{lev}}} :J^2:, \quad c = 1.$$

The all-spin $\mathcal{W}_{1+\infty}$ vertex algebra requires the completed Drinfeld double/Fock evaluation data in addition to this local abelian witness.

Proof. The gauge field $\mathcal{A} \in \Omega^{0,1}(\mathbb{C}^3, \mathfrak{gl}_N)$ expands along the normal fiber as $\mathcal{A}(z, w_1, w_2) = \sum_{a,b \geq 0} J_{a,b}(z) w_1^a w_2^b d\bar{z}$, where $J_{a,b}$ are fields on $C = \mathbb{C}_z$. The lowest BRST class is the $\mathfrak{gl}_1$ current $J(z)$. The OPE is computed from the quadratic part of the 6d hCS action restricted to the formal neighborhood, and the normal equivariant weights enter through $\Psi_{\text{lev}} = -\sigma_2(h_1, h_2, h_3)$. The CY condition $h_1 + h_2 + h_3 = 0$ gives $\sigma_2 = -(h_1^2 + h_2^2 + h_3^2)/2$, so $\Psi_{\text{lev}} = (h_1^2 + h_2^2 + h_3^2)/2$. The Sugawara formula for a nonzero-level Heisenberg current gives $c = 1$. $\square$

Remark 3.4 (Level-prefixed r -matrix). The classical r -matrix of the $\widehat{\mathfrak{gl}}_1$ current subalgebra derived from the 6d reduction carries level $\Psi = -\sigma_2$, not a normalised level $k = 1$:

$$r^{\text{Heis}}(z) = \frac{\Psi}{z} = \frac{-\sigma_2(h_1, h_2, h_3)}{z}.$$

At $\Psi = 0$ the r -matrix vanishes and the Sugawara construction is singular; this is the degenerate locus sitting outside the non-critical Koszul locus. The averaging map gives $\text{av}(r(z)) = \Psi = \kappa_{\text{ch}}(H_\Psi)$. For the full $\mathcal{W}_{1+\infty}$ algebra, higher-spin r -matrix components enter through the universal Miura cross-term: the leading term $J \otimes W_{s-1}$ has coefficient $(\Psi - 1)/\Psi$ for every $s \geq 2$. The central charge $c = 1$ is the rank of the underlying free boson and is independent of Ψ ; Prochazka–Rapcak parametrise $\mathcal{W}_{1+\infty}$ by $(\lambda_1, \lambda_2, \lambda_3)$ with $\sum_i 1/\lambda_i = 0$. The normalisation $\Psi = -\sigma_2$ is the one for which the equivariant hCS quadratic form gives $J(z)J(w) \sim \Psi/(z-w)^2$.

3.3 Defect algebras for general CY_3

For a general CY_3 -fold Y with a curve $C \subset Y$, the defect algebra depends on the normal bundle $N_{C/Y}$. The normal bundle is a rank-2 vector bundle on C ; its splitting type and degree control the defect algebra.

Example 3.5 (Conifold). For the resolved conifold $Y = \text{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1)$ and $C = \mathbb{P}^1$ (the zero section): $N_{C/Y} = \mathcal{O}(-1)^{\oplus 2}$. The defect algebra on C has $\kappa_{\text{ch}} = 1$ (from the DT computation) and shadow class G (shadow depth 2). The negative degrees of the normal bundle constrain the spin content: only finitely many spins contribute, and the resulting algebra is a truncation of $\mathcal{W}_{1+\infty}$.

Example 3.6 (Local $\mathbb{P}^2$). For $Y = \text{Tot}(K_{\mathbb{P}^2} \rightarrow \mathbb{P}^2)$ and a line $C = \mathbb{P}^1 \subset \mathbb{P}^2$: $N_{C/Y} = \mathcal{O}(1) \oplus \mathcal{O}(-3)$. The defect algebra has $\kappa_{\text{ch}} = 3/2$ and shadow class M (infinite shadow depth). The mixed splitting type of the normal bundle introduces asymmetry between the two spectral parameters.

4 The CoHA as ordered Hall input

4.1 Critical cohomological Hall algebras

For a quiver with potential (Q, W) arising from a toric CY_3 -fold, the critical CoHA is the $\mathbb{Z}$ -graded associative algebra

$$\text{CoHA}(Q, W) = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\text{BM}}(\text{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ is the vanishing cycle sheaf and the multiplication comes from Hall-algebra extension of representations. The grading by dimension vectors $\mathbf{d}$ is the CY analogue of the weight-space decomposition of a quantum group.

Theorem 4.1 (Schiffmann–Vasserot; Rapcak–Soibelman–Yang–Zhao). *For a toric CY_3 X without compact 4-cycles:*

$$\text{CoHA}(Q_X, W_X) \simeq Y^+(\widehat{\mathfrak{g}}_{Q_X})$$

where $Y^+(\widehat{\mathfrak{g}}_{Q_X})$ is the positive half of the affine super Yangian attached to the toric quiver. For $\mathbb{C}^3$ (Jordan quiver, cubic potential): $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$. The $\mathcal{W}_{1+\infty}$ vertex realisation appears only after passing to the Drinfeld double, or to the corresponding evaluation shadow.

4.2 The CoHA carries the positive E_1 half

The bare critical CoHA is the positive half of the CY quantum group. It supplies the ordered Hall product. Spectral coproducts, Hopf pairings, antipodes, Cartan currents, and negative roots belong only after localization, completion, and passage to the Drinfeld double.

Proposition 4.2 (CoHA positive-half data). *The critical CoHA $\text{CoHA}(Q, W)$ of a toric CY_3 carries the positive E_1 data:*

- (H1) Ordered Hall product. *The Hall multiplication (extension of short exact sequences $\circ \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow \circ$ with sub before quotient) is an ordered (E_1) product. The ordering is physical: the dimension-vector filtration imposes a preferred direction.*
- (H2) Positive half. *For the Jordan quiver, this ordered product identifies $\text{CoHA}(\mathbb{C}^3)$ with $Y^+(\widehat{\mathfrak{gl}}_1)$.*
- (H3) Double input. *A spectral coproduct or Hopf antipode requires the opposite half, Cartan sector, Hopf pairing, and completed Drinfeld double. These structures are extra data beyond the bare positive CoHA.*

Remark 4.3 (Hopf data require the double). The transfer-matrix antipode $S(T(u)) = T(u)^{-1}$ is a statement in the completed Yangian or toroidal double. The Hopf identity requires the negative half and the Cartan current in addition to the positive CoHA.

4.3 The R -matrix and the Drinfeld coproduct

The R -matrix of the CoHA arises from the Ext-quiver: the extension groups $\text{Ext}^1(V', V'')$ between representations determine the poles of the R -matrix, and the CY condition (self-duality of the Ext-quiver) constrains the R -matrix to satisfy the Yang–Baxter equation.

Proposition 4.4 (Universal Miura coefficient). *The Drinfeld coproduct Δ_z for $\mathcal{W}_{1+\infty}$ has the universal Miura coefficient $(\Psi - 1)/\Psi$ on the leading cross-term $J \otimes W_{s-1} + W_{s-1} \otimes J$ at every spin $s \geq 2$. At spin 2:*

$$\Delta_z(T) = T \otimes 1 + 1 \otimes T + \frac{\Psi - 1}{\Psi} J \otimes J + O(z^{-1}).$$

The coefficient $(\Psi - 1)/\Psi$ is structural: it arises from the Prochazka–Rapkak Miura factorization $T(u) = \prod_i (u + \Delta_i)$ in three steps: (1) single- J sector gives $1/\Psi$; (2) Drinfeld shift adds $+1$; (3) lower sectors do not contribute. The coefficient persists at all spins $s \geq 2$ with direct symbolic expansion through spins 2–6 giving the same coefficient.

Remark 4.5 (Coincidental agreement at $\Psi = 2$). At $\Psi = 2$: $(\Psi - 1)/\Psi = 1/2 = 1/\Psi$. The two expressions $(\Psi - 1)/\Psi$ and $1/\Psi$ agree at this single point, which is the self-dual level. The distinction is visible at $\Psi = 3$: $(\Psi - 1)/\Psi = 2/3 \neq 1/3 = 1/\Psi$. Verify at three or more parameter values to distinguish the two.

5 Quantum toroidal algebras and the CY condition

The quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ is the two-parameter deformation of the double current algebra $\mathfrak{gl}_1 \otimes \mathbb{C}[u^{\pm 1}, v^{\pm 1}]$, carrying the full $\mathbb{Z} \times \mathbb{Z}$ bigrading from the two loop directions.

5.1 The CY parameter constraint

In the additive parametrization $q_i = e^{h_i}$, the three equivariant parameters of the Ω -background satisfy the CY condition:

$$h_1 + h_2 + h_3 = 0 \quad \text{equivalently} \quad q_1 q_2 q_3 = 1.$$

Two independent parameters survive. The standard (q, t) -parametrization identifies $q_1 = q$, $q_2 = t^{-1}$, $q_3 = t/q$.

Definition 5.1 (CY reflection identity). The *trigonometric structure function* of the quantum toroidal algebra is

$$G(x; q_1, q_2, q_3) = \frac{(1 - q_1 x)(1 - q_2 x)(1 - q_3 x)}{(1 - q_1^{-1} x)(1 - q_2^{-1} x)(1 - q_3^{-1} x)}.$$

The *CY reflection identity* is

$$G(x) \cdot G(1/x) = 1.$$

Proposition 5.2 (CY reflection characterizes the CY condition). *Writing $q_1 q_2 q_3 = \omega$, one has*

$$G(x)G(x^{-1}) = \omega^2.$$

Thus the connected exponential Calabi–Yau branch $q_1 q_2 q_3 = 1$ implies the reflection identity $G(x)G(x^{-1}) = 1$. Algebraically the branch $\omega = -1$ also satisfies the squared identity; the converse requires the chosen connected/formal branch.

Proof. Direct computation. The numerator of $G(x) \cdot G(1/x)$ is $\prod_i (1 - q_i x)(1 - q_i/x) = \prod_i (1 - q_i x)(x - q_i)/x$. The denominator is $\prod_i (1 - q_i^{-1} x)(1 - q_i^{-1}/x) = \prod_i (1 - q_i^{-1} x)(x - q_i^{-1})/x$. The ratio simplifies to $\prod_i (q_i/q_i^{-1}) = \prod_i q_i^2 = (q_1 q_2 q_3)^2$. $\square$

5.2 The Miki automorphism

The Miki S_3 symmetry is an automorphism of the completed quantum toroidal/DIM algebra. On parameters it permutes (q_1, q_2, q_3) , but the automorphism is more than a cyclic relabelling of the positive half. It uses the completed algebra, Cartan sector, and opposite directions.

Remark 5.3 (The Miki automorphism and the operadic action). The E_3 -operad acts on the derived center $Z_{\text{ch}}^{\text{der}}(\mathcal{A}_C)$ through the higher Deligne conjecture. The Miki automorphism acts on the quantum toroidal algebra itself, permuting the three equivariant directions. These are different actions at different categorical levels: the E_3 -action is on the algebraic closed-sector sector $Z_{\text{ch}}^{\text{der}}(\mathcal{A}_C)$, while a physical-bulk reading requires the OCA/HT comparison datum. The Miki automorphism is on the boundary algebra (the chiral algebra itself). The relationship between them is mediated by the E_n operadic circle, but the Miki automorphism requires the completed toroidal algebra.

5.3 The DDCA–toroidal bridge

The deformed double current algebra $\text{DDCA}_k(\mathfrak{gl}_K) = U(\mathfrak{gl}_K \otimes \mathbb{C}[u, v])$ carries two spectral parameters from the two holomorphic directions of $\mathbb{C}^2$ in the 5d theory. The quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_K)$ carries two deformation parameters from the two loop directions.

Conjecture 5.4 (DDCA–toroidal bridge). *Writing $q = e^{\hbar_1}$, $t = e^{\hbar_2}$, the DDCA is the rational $(\hbar_1, \hbar_2 \rightarrow 0)$ degeneration of the quantum toroidal algebra in the category of filtered algebras. The two one-parameter degenerations produce a commuting square:*

$$\begin{array}{ccc} U_{q,t}(\widehat{\mathfrak{gl}}_K) & \xrightarrow{t \rightarrow 1} & Y(\widehat{\mathfrak{gl}}_K) \otimes U(\mathbb{C}[v]) \\ \downarrow q \rightarrow 1 & & \downarrow \hbar_1 \rightarrow 0 \\ U(\mathbb{C}[u]) \otimes Y(\widehat{\mathfrak{gl}}_K) & \xrightarrow{\hbar_2 \rightarrow 0} & \text{DDCA}_k(\mathfrak{gl}_K) \end{array}$$

The diagonal $\hbar_1 = \hbar_2 \rightarrow 0$ recovers the affine Yangian $Y(\widehat{\mathfrak{gl}}_K)$ as the common degeneration. The modular characteristic is continuous across the bridge: $\kappa_{\text{ch}}(\text{DDCA}_k(\mathfrak{gl}_K)) = K(k + K)/2 = \lim_{\hbar_i \rightarrow 0} \kappa_{\text{ch}}(U_{q,t})$.

6 The Drinfeld center closes the E_n circle

The Drinfeld center $\mathcal{Z}(\text{Rep}^{E_1}(A))$ is the universal braided monoidal category receiving a central monoidal functor from $\text{Rep}^{E_1}(A)$. For the CoHA of a toric CY_3 -fold:

Theorem 6.1 (Drinfeld center for constructed CY_3 chiral loci). *For A_C in the constructed BZFN/CoHA loci of a toric CY_3 -fold X :*

$$\mathcal{Z}(\text{Rep}^{E_1}(A_C)) \simeq \text{Rep}^{E_2}(Z_{\text{ch}}^{\text{der}}(A_C))$$

where $Z_{\text{ch}}^{\text{der}}(A_C) = C_{\text{ch}}^{\bullet}(A_C, A_C)$ is the chiral derived center. The forgetful functor $U: \mathcal{Z}(\text{Rep}^{E_1}(A_C)) \rightarrow \text{Rep}^{E_1}(A_C)$ is the underlying-monoidal functor; its right adjoint, the Drinfeld center, equips objects with half-braidings. This is the categorical-centre step, not the lossy averaging map to the scalar shadow.

Remark 6.2 (Ordinary center and derived center). For $A = H_k$ (Heisenberg at level k): the ordinary center $Z(A) = \mathbb{C}$ has dimension 1; the derived center $Z_{\text{ch}}^{\text{der}}(A)$ has dimension 3, capturing Ext^1 and Ext^2 invisible to the commutant. The five invariants matching at six levels (conjectured for Heisenberg; proved end-to-end for $\mathbb{C}^3$ via the CoHA) confirm that the Drinfeld center of the E_1 -monoidal representation category is the correct categorification of the chiral derived center.

The passage $E_1 \rightarrow E_2$ via the Drinfeld center is the CY incarnation of the E_1 -to- E_2 lift of Volume I. The averaging map $\text{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\text{mod}}$ projects the ordered R -matrix to the scalar κ_{ch} ; the Drinfeld center is the categorical analogue, recovering the R -matrix data as the braiding on the center category.

Direct restriction from E_d to E_2 (for $d \geq 3$) gives only symmetric braiding:

$$\pi_1(\text{Conf}_2(\mathbb{R}^d)) = 0, \quad \pi_1(\text{Conf}_2(\mathbb{R}^d)/\mathfrak{S}_2) = \mathbb{Z}/2 \quad (d \geq 3).$$

The Drinfeld center is the source of the nontrivial centre-side half-braiding for CY_3 quantum groups.

7 Shadow = GW($\mathbb{C}^3$) and wall-crossing

7.1 The shadow tower as Gromov–Witten theory

At $\kappa_{\text{ch}} = \Psi$ (the deformation parameter of the low-spin defect channel), the shadow obstruction tower recovers the genus-1 constant-map contribution for $\mathbb{C}^3$. The all-genus virtual identification requires the uniform-weight and $\Phi_3^{(\Sigma_2, C)}$ hypotheses; the motivic identification remains conjectural.

Proposition 7.1 (Genus-one shadow and the DT character of $\mathbb{C}^3$). *At genus 1, the shadow specialization $\kappa_{\text{ch}} = \Psi$ matches the constant-map/DT genus-one channel for $\mathbb{C}^3$. On the DT side, the MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ encodes the partition function as the second-quantized bar/DT character in the non-compact degree-zero normalization.*

The identification shadow = GW operates at three levels:

- (i) *Genus-1*: the shadow Eisenstein theorem $D_2(\mathcal{A}, s) = -24\kappa_{\text{ch}} \cdot \zeta(s)\zeta(s-1)$ (Volume I) gives the genus-1 DT partition function via the Borcherds lift.
- (ii) *Virtual*: under the uniform-weight and $\Phi_3^{(\Sigma_2, C)}$ hypotheses, the shadow coefficient S_r at degree r matches the virtual Euler characteristic of the moduli of r -dimensional sheaves.
- (iii) *Motivic*: the full chiral quantum group structure (R -matrix, coproduct, RTT) is conjecturally matched with the motivic DT invariants of Kontsevich–Soibelman.

7.2 Wall-crossing as MC gauge equivalence

For a toric CY_3 with fan Σ , Bridgeland stability conditions partition the space of central charges into chambers. Each chamber σ determines a quiver presentation (Q_σ, W_σ) and a CoHA $\text{CoHA}(Q_\sigma, W_\sigma)$.

Conjecture 7.2 (Wall-crossing = MC gauge equivalence). *Wall-crossing between stability chambers σ and σ' is Maurer–Cartan gauge equivalence in the modular convolution algebra: $\Theta_\sigma \sim \Theta_{\sigma'}$. The Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation. The global chiral algebra is the homotopy colimit over the chamber decomposition: $A_C = \text{hocolim}_\sigma A_{Q_\sigma}$.*

For $\mathbb{C}^3$ (single chamber): the conjecture is vacuous. For the resolved conifold (two chambers): the wall-crossing formula is the flop relation on DT partition functions (Bryan–Steinberg), and the MC gauge equivalence is the bar-cobar transport across the flop.

Remark 7.3 (The four faces of wall-crossing). Wall-crossing admits four descriptions in the programme:

- (i) *DT/BPS*: change of BPS spectrum across a wall of marginal stability.
- (ii) *MC gauge*: the universal Maurer–Cartan element Θ_A transforms by gauge equivalence in the modular convolution algebra.
- (iii) *Scattering diagram*: the Gross–Siebert consistency condition IS the E_1 Maurer–Cartan equation (proved for affine KM).
- (iv) *Root system*: reflections in the Weyl group of the BKM superalgebra $\mathfrak{g}_X$.

The identification of (ii) with (iii) is proved; the identification with (i) and (iv) is conjectural in the CY_3 setting.

8 The CY_4 obstruction and the p_1 -twisted double current algebra

The fourth-dimensional stage of the holomorphic Chern–Simons hierarchy is obstructed. The untwisted quantum group associated to a CY_4 geometry is replaced by a p_1 -twisted double current algebra: the first Pontryagin class of the tangent bundle appears as a cocycle on the Drinfeld double.

8.1 The obstruction in $\pi_4(BU)$

Theorem 8.1 (CY_4 first obstruction). *Let X be a compact CY_4 manifold. The obstruction to promoting $\Phi(D^b \text{Coh}(X))$ to an E_4 -algebra lies in $\pi_4(BU) \cong \mathbb{Z}$, realised as the framing class of the stable normal bundle to the cyclic trace on X .*

Proof. By Bott periodicity, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The equality $\pi_3(BU) = 0$ follows from $\pi_2(U) = 0$. The E_1 -direction framing of the cyclic trace on X carries a characteristic obstruction class in $H^4(X; \mathbb{Z})$, represented by $p_1(T_X)$. Scalar obstructions are obtained by pairing this class with specified four-cycles, or by the degree-compatible invariant $\int_X p_1(T_X)^2$ on the real eight-manifold X . Obstruction theory for E_1 -to- E_4 promotion locates this first class in the $\pi_4(BU)$ stage. Its vanishing removes only this obstruction; a genuine E_4 lift still requires the higher obstruction tower and holomorphic-twist, framing, and completion witnesses. $\square$

Remark 8.2. Contrast CY_3 : there $\pi_3(BU) = 0$, so this first scalar obstruction vanishes. The CY_4 case is the first step where this homotopy group is non-trivial; the full higher obstruction tower remains separate.

8.2 The Drinfeld double presentation

Definition 8.3 (p_1 -twisted double current algebra). Let X be a CY_4 and $\mathfrak{g}$ a simple Lie algebra. Let $Y^+(\mathfrak{g})$ and $Y^-(\mathfrak{g})$ be the positive and negative half-Yangians (in the Drinfeld realisation: Chevalley generators $\{x_i^{\pm, r}\}_{r \geq 0}$ subject to the Yangian relations over $\mathbb{C}[\hbar]$). The p_1 -twisted double current algebra associated to X is the Drinfeld double

$$Y_{p_1}(X) := Y^+(\mathfrak{g}) \bowtie_c Y^-(\mathfrak{g})$$

equipped with the cocycle

$$c(x, y) = \frac{1}{24} \left\langle x \cup y \cup p_1(T_X), [X] \right\rangle, \quad x, y \in H^\bullet(X; \mathfrak{g}),$$

where $p_1(T_X) \in H^4(X; \mathbb{Z})$ is the first Pontryagin class and $[X] \in H_8(X; \mathbb{Z})$ the fundamental class.

Remark 8.4. The algebra $Y_{p_1}(X)$ is a p_1 -twisted Drinfeld double. Its cocycle depends on X through p_1 , and its RTT presentation contains a genuine $O(\hbar^2)$ correction absent from ordinary Yangians; the notation “ CY_4 Yangian” suppresses the obstruction.

8.3 The RTT relation through $O(\hbar^2)$

Conjecture 8.5 (p_1 -twisted RTT). *Let*

$$N_{p_1^2}(X) := \int_X p_1(T_X)^2.$$

The R -matrix of $Y_{p_1}(X)$ satisfies

$$R_{p_1}(u) = 1 + \frac{\hbar P}{u} + \hbar^2 \frac{N_{p_1^2}(X) \Omega_{\text{Cartan}}}{u^2} + O(\hbar^3),$$

where P is the permutation operator and Ω_{Cartan} the Cartan component of the quadratic Casimir.

Remark 8.6 (Proof obligation). The displayed $O(\hbar^2)$ term requires a deformation-complex calculation for the p_1 -twisted double, or a primary source proving the same RTT normalisation. Whitehead's lemma supplies only the cohomological vanishing input.

8.4 CY_4 twisted-supergravity analogue

Conjecture 8.7 (CY_4 twisted-supergravity boundary algebra). *The boundary algebra of the conjectural CY_4 twisted-supergravity analogue on $X \times \mathbb{R}$, with X a CY_4 , is the p_1 -twisted double current algebra $Y_{p_1}(X)$.*

Remark 8.8 (Evidence). The statement is the CY_4 analogue of the 5d hCS/Yangian-VOA theorem on $\mathbb{R} \times \mathbb{C}^2$ and of the 6d hCS framework of Section 2. Since a CY_4 has real dimension 8, the space $X \times \mathbb{R}$ is 9-dimensional. The statement is a CY_4 twisted-supergravity conjecture. The cocycle c is predicted by the one-loop anomaly, proportional to $p_1(T_X)$.

8.5 Comparison with Maulik–Okounkov

Proposition 8.9 (Maulik–Okounkov comparison). *At $X = \mathbb{C}^4$, the twist c degenerates to the equivariant Omega-background of Maulik–Okounkov, and the specialisation map*

$$Y_{p_1}(\mathbb{C}^4) \longrightarrow U_{q,t_1,t_2}(\widehat{\mathfrak{gl}}_1)$$

identifies the p_1 -twisted double current algebra with the quantum toroidal $\mathfrak{gl}_1$ algebra of Maulik–Okounkov.

Remark 8.10 (Attribution). Cited from Maulik–Okounkov, *Quantum groups and quantum cohomology*, Astérisque 408 (2019), and Schiffmann–Vasserot. The vanishing of $N(\mathbb{C}^4)$ on the non-compact target is cancelled by the equivariant integration that produces the (q, t_1, t_2) parameters; this is the standard refinement dictionary.

8.6 p_1 -twisted stratum of $K_3 \times K_3$

Proposition 8.11 ($K_3 \times K_3$ is p_1 -twisted). *For $X = K_3 \times K_3$,*

$$N_{p_1^2}(X) = \int_{K_3 \times K_3} p_1(T_X)^2 = 4608,$$

and $p_1(T_X) \neq 0$. Thus $K_3 \times K_3$ lies on the p_1 -twisted stratum. The untwisted Drinfeld double and pointwise E_4 lift require separate vanishing data.

Proof. The Pontryagin class splits: $p_1(T_{K_3 \times K_3}) = \pi_1^* p_1(T_{K_3}) + \pi_2^* p_1(T_{K_3})$, where π_i are the two projections. For a K_3 surface $p_1(T_{K_3}) = -2c_2(T_{K_3}) = -48 [\text{pt}]$ in $H^4(K_3)$. Therefore

$$p_1(T_{K_3 \times K_3}) = -48(\pi_1^* [\text{pt}] + \pi_2^* [\text{pt}]).$$

Squaring gives

$$p_1(T_{K_3 \times K_3})^2 = 2 \cdot 48^2 \pi_1^*[\text{pt}] \pi_2^*[\text{pt}],$$

and integration over $K_3 \times K_3$ gives $2 \cdot 48^2 = 4608$. $\square$

Remark 8.12. The product $K_3 \times K_3$ is off the S^4 -framed stratum. Its natural CY_4 quantum group is p_1 -twisted.

9 Codimension-2 defects, CoHA, and the toroidal formal disk

The 6d holomorphic Chern–Simons hierarchy has three load-bearing tests: the codimension-2 defect algebra on a curve $C \subset \mathbb{C}^3$, the identification $\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ as the *positive half* of $\mathcal{W}_{1+\infty}$, and the formal-disk specialisation of the toroidal algebra via the Schiffmann–Vasserot CoHA and the Miki *DIM* presentation.

Theorem 9.1 (Codim-2 low-spin defect at non-critical level). *Let $Y = \mathbb{C}^3$ with holomorphic volume form $\Omega_Y = dz_1 \wedge dz_2 \wedge dz_3$ and let $C = \{z_2 = z_3 = 0\} \subset Y$ be a codim-2 holomorphic curve. The 6d holomorphic Chern–Simons defect on C has a Heisenberg current with level*

$$\Psi_{\text{lev}} = -\sigma_2(h_1, h_2, h_3),$$

where h_1 is the tangent weight and (h_2, h_3) are the normal weights, subject to $h_1 + h_2 + h_3 = 0$. The Sugawara stress tensor has central charge $c = 1$.

Proof. Reduction of 6d hCS along $N_{C/Y}$ is a codim-2 boundary condition; the descent of the gauge algebra from $\text{hol}(Y)$ to $\text{hol}(C)$ factors through the normal Koszul complex, producing a chiral extension of $\widehat{\mathfrak{gl}}_1$. The equivariant normal parameters are h_2, h_3 and the tangent weight is h_1 . Their elementary symmetric polynomial σ_2 gives the Heisenberg level $\Psi_{\text{lev}} = -\sigma_2$. The vanishing $\sigma_1 = 0$ is the CY condition. Non-criticality $\Psi_{\text{lev}} \neq 0, \infty$ places the Heisenberg current on the Sugawara locus. $\square$

Theorem 9.2 (CoHA($\mathbb{C}^3$) as the positive half). *The critical cohomological Hall algebra of $\mathbb{C}^3$ (= Jordan quiver with cubic potential) is isomorphic to the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$ of the affine Yangian of $\widehat{\mathfrak{gl}}_1$:*

$$\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1).$$

The full $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is recovered as the Drinfeld double $Y^+ \bowtie Y^-$, with the opposite half Y^- realised either by the opposite orientation $(\mathbb{C}^3)^{\text{op}}$ (conjugate quiver with potential $-W$) or by a Koszul dual CoHA¹ construction.

Proof. Schiffmann–Vasserot prove $\text{CoHA}(Q, W) \simeq Y^+$ as the Hopf algebra generated by simple-object fundamental classes, with Hall multiplication for positive roots; the shuffle presentation exhibits only Y^+ . The full Yangian with Cartan currents and negative roots requires either a second copy via $(\mathbb{C}^3)^{\text{op}}$ or the derived Drinfeld double. Thus the precise identification is $\text{CoHA} \simeq Y^+$. Character coincidence with $\mathcal{W}_{1+\infty}$ is weaker than a bar equivalence. $\square$

Theorem 9.3 (Toroidal formal-disk identification). *On the formal disk $D = \text{Spec } \mathbb{k}[[z]]$, the completed quantum toroidal algebra $U_{q,t}(\widehat{\mathfrak{gl}}_1)$ carries the structure of an E_1 -chiral quantum group: (i) the completed Miki S_3 -automorphism realizes the Ω -background Weyl symmetry on the branch $q_1 q_2 q_3 = 1$; (ii) the*

spectral R -matrix from the completed SV /toroidal structure satisfies the additive Yang–Baxter equation; (iii) the formal-disk chiral quantum-group equivalence holds at that level with associator dependence controlled by the Etingof–Kazhdan functor on the Koszul locus.

Remark 9.4 (Global toroidal: conditional). The extension of Theorem 9.3 from the formal disk to a global curve $\mathbb{P}^1 \times \mathbb{P}^1$ is conditional on chain-level topologization of the underlying class M chiral algebra: the formal-disk statement is unconditional, the global statement inherits the class M chain-level topologization hypothesis and the modular-family base-change problem (proved cohomologically; the chain-level statement depends on Francis–Gaitsgory base change for the relative Ran prestack).

Remark 9.5 (Geometric vs algebraic model discipline). The parameters (h_1, h_2, h_3) are formal algebraic variables in $\text{End}_{\mathcal{A}}^{\text{ch}}$. Fulton–MacPherson variables on $\text{FM}_k(\mathbb{C})$, the Hopf-algebra identity $\text{CoHA}(\mathbb{C}^3) \simeq Y^+$, and BZFN centre comparisons live in three distinct ambients: configuration-space geometry, algebraic Hopf algebras, and the pair $(\mathcal{A}^{\text{ch}}, \mathcal{A}_{\text{mode}})$.

Remark 9.6 (Proof ambients). Theorem 9.1 rests on the formal BRST calculation at non-critical level (proved cohomologically for affine KM; class M cases via DS reduction conditional). Theorem 9.2 uses the Schiffmann–Vasserot positive-half normalization. Identifying the CoHA with $\mathcal{W}_{1+\infty}$ gives at most a character identity until the Drinfeld double is included. Theorem 9.3 is proved at the formal disk; global extension conditional per Remark 9.4.

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